



Pau Juan-García

Principal Consultant



PROFESSION

Principal Scientist

PROFESSIONAL ASSOCIATIONS

C. WEM

QUALIFICATIONS

BSc, MSc, PhD (Hons)

JOINED ATKINSRÉALIS

2015

NATIONALITY / LANGUAGES

Spanish / Spanish, English,
Catalan, French, Chinese (basic)

PROFILE

Pau is a Principal Environmental Consultant in the Water Management practice at AtkinsRéalis and a recognised technical leader in water-quality modelling and catchment planning. He is the lead developer of SAGIS (UKWIR), the primary water-quality catchment planning tool used by environmental regulators in the UK, and a Defra Business-Critical Model (BCM). He also leads development of the River Quality Planning (RQP) tool for the Environment Agency. Together, these tools support regulatory decision-making and the allocation of millions of pounds of investment across AMP cycles.

Pau has over 12 years' experience in the water sector; delivering water-quality modelling, source-apportionment assessments, and GIS-based environmental analysis. He has acted as technical lead for tools and modelling solutions supporting catchment management, nutrient mitigation, climate-change impact assessment and permit optioneering. His technical skillset includes statistical and machine-learning methods, and the analysis of large geospatial datasets to inform water policy and infrastructure planning. He has worked closely with regulatory bodies including the Environment Agency and SEPA. Pau is a Marie Curie laureate and holds an international PhD in Water Science and Technology, an MSc in Environmental Engineering, and a BSc in Environmental Science. He is a Chartered Environmental Manager (C.WEM) with CIWEM, has served on scientific committees for international conferences, and has authored multiple peer-reviewed publications. He is also an experienced project manager, information manager and line manager

KEY EXPERIENCE

- Water Quality Modelling
- Software development (Python, C#, R, ArcGIS)
- Data analysis and visualisation
- Machine learning and AI
- Nutrient mitigation
- Climate change impact assessment

**EXPERIENCE WITH
ATKINSRÉALIS
(2015 - 2026)**

- **2026 UKWIR SAGIS Agriculture data update (project manager)**
Project manager for a SAGIS project with UKWIR to update the agriculture and livestock datasets, ensuring data quality, consistency, and suitability for regulatory and planning use.
- **2026 Environment Agency River Quality Planning software tool development - phase 2 (lead developer)**
Lead developer in a project with the Environment Agency to continue enhancing the RQP software package by introducing new capabilities ammonia modelling, sensitivity analysis and metal permitting.
- **2025 UKWIR SAGIS Support 2025/2026 (lead developer)**
Lead developer providing ongoing technical support and enhancements to the SAGIS platform for UKWIR, ensuring reliable performance and continued alignment with regulatory and industry needs during 2025/2026
- **2025 Welsh Water – DWMP Coastal Lead (water quality expert)**
Water quality expert leading the coastal elements of Welsh Water's Drainage and Wastewater management plan (DWMP), providing technical direction on the water quality risk assessment methodology, evidence development, and regulatory alignment with Ofwat.
- **2025 Anglian Water – WRC Capacity Review (project manager)**
Project manager leading the development of a tool to deliver the DWMP Sewage Treatment performance indicator for AWS. In charge of coordinating technical inputs, stakeholder engagement, and delivery of evidence to support investment and planning decisions.
- **2025 Updating Scottish SAGIS models (technical lead)**
Technical lead overseeing updates to Scottish SAGIS models, providing expert direction on model development, data integration, and assurance of technical robustness for regulatory and planning applications.
- **2025 Chemical Investigation Program (CIP) 4 (lead data analyst)**
Lead data analyst for CIP4, leading the analysis, quality assurance, and interpretation of chemical monitoring data to support regulatory compliance and investigative reporting.
- **2025 Lady's Island Lagoon (technical expert)**
Technical expert providing specialist input on water quality and environmental assessment for the Lady's Island Lagoon project in Ireland.
- **2025 Nitrate Technically Achievable Limit Trials (data analyst)**
Data analyst in a project with UKWIR to generate robust evidence on nitrate removal performance across wastewater treatment processes to inform a new Technically Achievable Limit (TAL).
- **2025 Defra Assessing ammonia error propagation (technical specialist)**
Lead modeller in a project with Defra to assess the reliability of ammonia concentration estimates in rivers by analysing water quality data from selected monitoring sites, using a Monte Carlo sensitivity analysis.

- **2025 SEPA Updating SAGIS models (technical lead)**
Lead specialist in a project with SEPA to convert and update all SAGIS models to the newest version in ArcGIS Pro 3.3 from its current version in ArcMap.
- **2025 A nutrient management plan for the Usk catchment (project manager)**
Project manager in a project to design the Usk Catchment Nutrient Management Plan. Using NFM Studio, Sediment Studio, and FARMSCOOPER, we modelled runoff, erosion, and pollution from various sources in the Usk Catchment. An interactive GIS map was built to highlight high-risk zones and recommends nature-based solutions to improve water quality and support development.
- **2025 UKWIR SAGIS Training Videos (technical specialist)**
Lead specialist in a project with UKWIR. To further support the use of SAGIS, we produced training materials in the form of video tutorials and guided exercises to train the next generation of modellers and water quality planners.
- **2024 UKWIR Real-time Bathing Water notifications AI (technical specialist)**
Lead specialist in machine learning in an innovative UKWIR project that aims to bring together new technological developments and collaboration with engaged stakeholders, aiming to revolutionise the way the public is informed about bathing water quality. The project work includes a comprehensive literature review and horizon scanning of the best global practice, including interviews with technical specialists from across the world. This is followed by deterministic bathing water quality modelling at 10 diverse locations across the UK to inform a standard modelling approach, as well as a review of sampling requirements to inform new technologies and machine learning.
- **2024 SIMCAT RQP-MPER Open Source (technical specialist)**
Lead specialist in a project with the Environment Agency to help them set up online GitHub repositories for two software (SIMCAT and RQP-MPER) and build a development community of users around them.
- **2024 Environment Agency PFAS investigation (technical specialist)**
I oversaw my team with data management, download and analysis tasks on a national scale investigation to look at PFAS sources across the UK.
- **2024 SAGIS development support (lead developer)**
I am the lead developer in a support contract with UKWIR to continue to develop and improve the SAGIS water quality modelling system, responsible for millions of pounds of investment in each AMP cycle.
- **2023-2024 State of the Nation coastal modelling (technical specialist)**
I am in charge of the development and correct implementation of the script used to automate the State of the Nation coastal modelling protocol designed by the Environment Agency in a series of investigations.
- **2023 Wessex Water AMP8 SAGIS support (secondment)**
As part of this project, I was seconded to Wessex Water for 9 months. During this time, I lead my team to carry out a series of studies, including the analysis for several Environment Agency drivers: 'no deterioration', 'dry-weather flow exceedance' and

LTraCs and SSSI nutrient neutrality. I also studied the effects of climate change on these drivers and provided training in SAGIS to the company's employees.

- **2023 South Staffs CEE Climate change analysis (technical specialist)**
I was a technical specialist on this study for South Staffs that designed a new methodology to assess the effects of climate change on abstractions
- **2023 South East Water Broad Oak reservoir study (technical specialist)**
I was a technical specialist to carry out a scoping study of water quality pressures and design of a water quality sampling campaign for the scoping and design of the Broad Oak reservoir.
- **2023 A pollution inventory tool for the Irish (technical specialist)**
I was the lead expert on setting up UKWIR's Pollution Inventory tool for Irish Water to be used for reporting to the European Pollutant Release and Transfer Register.
- **2023 United Utilities Tabley Mere water quality study (water quality modeller)**
I was the lead expert in a SAGIS modelling study to assess the effect of a CSO to the Tabley Mere Lake for United Utilities.
- **2023 United Utilities North-West Transfer Modelling (water quality modeller)**
I was a technical expert assisting in a SAGIS modelling study to assess the effect on water quality of the abstractions pertaining to the North-West Transfer project in United Utilities.
- **2022 Welsh Water SAGIS modelling upgrade (project manager)**
I was the project manager in a water quality review of the SAGIS models for the River Tywi, Eden, Gwyrfaï and Meirionnydd Oakwoods.
- **2022-2023 SIMCAT scoping (technical specialist)**
Technical expert in a project to identify and appraise potential options for the ongoing management of the SIMCAT codebase. I carried out a desk-based study of open-source management practices, and a comparison of approaches adopted for other relatable software tools. In a second stage of this project funded directly by the Environment Agency I provided a recommendation to make SIMCAT open source.
- **2021-2022 Environment Agency – Coastal modelling**
This project for the Environment Agency consists of a series of modelling studies for the Taw-Torridge, Penzance and Bude areas, using the "State of the nation" methodology coined by the Environment Agency. My role consisted of designing the implementation of said methodology and carrying it out using Python, MIKE21, SWAN, XBeach and other modelling tools.
- **2021-2022 Wessex water modelling support for rCSMG (water quality modeller)**
Water quality modeller in study of phosphorus sources to identify the discharge permit criteria required to address DCWW's estimated 'fair share'. Key tasks include updating, designing scenarios, and calibrating the Wessex Water's water quality models. Other tasks include supporting Wessex Water in the undertaking of further simulation work.

- **2021-2022, UKWIR, Pollution Inventory Tool (Project manager and technical specialist)**

The objectives of this project were threefold: i) to provide an updated and revised version of the Pollution Inventory estimator tool to generate e-PRTR estimates of suitable credibility to regulators; ii) to demonstrate that the companies are using the best available information to support the requirements of regulators, and that this is provided on a consistent and comparable basis across the sector; and iii) to scope alternative platforms for the tool in future.
- **2021 Thames Water, Chalgrove STW (water quality modeller)**

Preparation of the solution development to Thames Water's Approval AMP7 Non-Infrastructure (NI) Project for the Chalgrove wastewater treatment works. I was the lead scientist and modeller in a study to assess the anticipated growth and to ensure the Water Framework Directive (WFD) requirements do not cause deterioration or prevent future improvement works to the Haseley Brook.
- **2021 Thames Water, SESRO (data engineer, modeller)**

Use of SAGIS-SIMCAT to undertake modelling for the planned South East Strategic Reservoir Option (SESRO) in Oxfordshire. The purpose of the SAGIS modelling is to assess the impact of the scheme on WFD compliance in the River Thames and risks to drinking water at downstream water treatment works.
- **2021 UKWIR, Fair Share (project manager and technical specialist)**

The aim of this project was to provide the water industry with the capability to rapidly undertake pollution allocation evaluations using SAGIS outputs. I helped design and tested an analysis tool that can rapidly evaluate the implications of different interpretations (or variations) of the Polluter Pays methodologies, at a variety of spatial scales, and for a range of determinands and supported the transfer of knowledge from the study to the wider SAGIS user community.
- **2021 UKWIR, SAGIS Support contract (lead software developer, modeller)**

Software developer in the maintenance and support of the SAGIS water quality simulation tool. The Source Apportionment Geographical Information System (SAGIS) is a digital information management and visualisation platform embedded within ArcGIS. In combination with the SIMCAT, SAGIS is used by Water Companies to support decision making as part of the Asset Management Planning (AMP) cycle process, and by Regulators for River Basin Management Planning. SAGIS is firmly established within the Water Industry's 'business-as-usual' planning practices. In the recent round of business planning for PR19, SAGIS outputs have served as a key evidence base for identifying wastewater treatment works requiring schemes to further control discharge quality. This support contract includes the development of new tools and functionality, as well as independent pieces of work such as creating a layer of targets and orthophosphate standards, and the development of scripts and data analysis to support SAGIS work.
- **2021 UKWIR, Chemical Investigation Programme (data analyst)**

I was responsible of the data analysis of water quality data samples across all water companies through Great Britain. The UKWIR Chemical Investigations Programme (CIP) is an ongoing research programme that addresses the likely implications of environmental legislation for the UK water industry. It has primarily been addressing the consequences of legislation relating to trace substances in the

water environment. It is coordinated by UK Water Industry Research as a collaborative programme involving the ten water companies in England and Wales.

- **2020-2021 EA AMR (geospatial data analyst)**
Relevant AMR datasets were identified within a recent EA/Defra report 'Framework for understanding environmental antimicrobial resistance in England'. I located and compiled selected datasets on sources of AMR into the environment into a structure of geodatabases. These data will be a foundation of further efforts on AMR. This project is a step towards the development of an understanding of the hazard characterisation of AMR in the environment. The overarching aim of the work is to bring together various datasets about potential sources of AMR in the environment into one database, as an easy-to-access resource.
- **2020 Anglian Water CSMG Investigations (water quality modeller)**
I modelled and designed measures to help achieve CSMG targets and favourable status in the catchment. The investigations followed a Judicial River and Consent Order that requires that the Environment Agency and Natural England to assess Natura 2000 sites and identify the measures necessary to achieve protected area objectives.
- **2020 SAGIS Climate change tool (technical specialist)**
Participated in the design and testing of a climate change and sensitivity analysis tool that enabled water companies and regulators to explicitly include climate change considerations in SAGIS-SIMCAT catchment and asset management planning applications.
- **2020-2021 Wessex water modelling support for PR24 (water quality modeller)**
I identified the discharge permit criteria required to address DCWW's estimated 'fair share'. Key tasks included updating, designing scenarios, and calibrating Wessex Water's SAGIS models. Other tasks include supporting Wessex Water in the undertaking of further simulation work.
- **2020 CDF-SW - DCS modelling - Taw Torridge (technical specialist)**
Working with the Environment Agency, I developed in-house tools to carry out the "State of the Nation" methodology for Coastal Modelling. This consists of a series of Python scripts that automate wave simulations in SWAN and MIKE21, as well as curve fitting to extrapolate simulation results to similar areas.
- **2019-2020 Clwyd SAGIS catchment investigation (water quality modeller)**
I applied a scenario modelling approach using the SAGIS-SIMCAT system to simulate the extent to which controls of phosphorous inputs from DCWW's assets might mitigate the risk of non-compliance within the Clwyd catchment and address DCWW's 'fair share'.
- **2019-2020 Upgrade of SAGIS Model Code (software developer)**
Software developer is key project for the water industry to migrate the SAGIS codebase to ArcGIS Pro. SAGIS was 'written' in VBA, which ceased to be available in newer versions of ArcGIS. This project created the new ESRI Add-In to install SAGIS in ArcGIS Pro using Python.
- **2018-2020 Extending and updating UKWIR's source apportionment tool; UKWIR, SEPA and Environment Agency (water quality modeller)**

Technical specialist responsible for the data analysis and appraisal of the update of the SAGIS-SIMCAT model to include faecal matter indicator organism (FIO) as a determinand and assess the capability of using SAGIS for FIO mitigation and decision making.

- **2018 Enhancing the Scottish SAGIS modelling platform, UKWIR and SEPA (water quality modeller)**

Technical specialist in the inclusion of lakes and reservoirs into SEPA's SAGIS models so that this offers an integrated catchment planning system.

- **2018 Extending and Updating UKWIR's SAGIS-SIMCAT decision support tool (software developer)**

Main software developer in project to expand SAGIS to include diffuse sources of pollution. The DST is an optimisation tool for water management built on top of the water quality modelling system SAGIS-SIMCAT.

- **2018 Extending and Updating UKWIR's source apportionment toolset (technical specialist)**

Technical specialist assisting on the deployment of a SAGIS-SIMCAT data analysis tool, consisting of an analytical methodology for detecting systematic bias and quantifying uncertainty in SAGIS input data. I developed said methodology in a Python modelling platform.

- **2018 Climate Change Adaptation and Resilient Cities: A Streamlined Approach for the Assessment of the Contribution to Climate Action for Multi-Sector Framework Loans (data analyst)**

Technical specialist to provide support on the data analysis of the European Investment Bank (EIB) Framework loan database. The analysis included identification of trends and production of summary statistics and a range of charts, figures, and visualisations.

- **2015-2018 Resilience of wastewater treatment to stress conditions. Marie Curie PhD grant (research scientist)**

Researcher and PhD candidate at TreatRec, a Marie Skłodowska-Curie Action European Industrial Doctorate program. The project arises from the collaboration between Academia (University of Girona, ICRA) and Industry (Atkins). The outputs include various publications and a PhD thesis to review resilience theory applied to wastewater treatment, and a framework for model-based resilience assessment and energy audit of STW using state of the art process modelling.